

SPEECH PROCESSING ARCHITECTURE SPECIFICATION (v1.0)

OneMeta S2S POC — Milestone 3 Foundation Freeze for Sub-400ms Target

Status: FROZEN

Target Latency: < 400ms

1. Objective & Target Latency Budget

Is specification ka maqsad Milestone 3 (Speech Processing, VAD, Chunking, Context Management) ke tamam engineering decisions aur interfaces ko freeze karna hai taaki final Speech-to-Speech (S2S) pipeline me **Sub-400ms TTFT/TTFA** target reliably achieve ho sake.

Pipeline Stage	Max Target Latency	Optimization Strategy
LiveKit Capture & Network Transport	20 ms	UDP / WebRTC low latency mode
Audio Resampling & PCM Standard Normalization	5 ms	C-based PyAV / zero-copy memoryview
VAD Frame Filtering & Speech Detection	10 ms	WebRTC VAD (Level 2/3), C-extension
Chunk Builder & Trigger Logic	15 ms	200ms silence or 1.2s hard cap trigger
Context Framing & Prompt Formatting	10 ms	Truncated sliding window (< 40 tokens)
Gemma 4 Inference Engine (TTFT)	180 ms	KV Cache reuse + Dynamic batching (Local GPU)
TTS Streaming Synthesis (TTFA)	100 ms	Token-by-token streaming audio synth
LiveKit Streaming Publish Back	20 ms	Reuse existing audio track
TOTAL PIPELINE TARGET	360 ms (< 400 ms)	End-to-End Budget Buffer = 40 ms

2. Audio Input Standard & Normalization

- **Sample Rate:** 16,000 Hz (16 kHz) (STT/VAD Models ke liye native)
- **Channels:** Mono (1 Channel)
- **Sample Format:** PCM 16-bit Signed integer (PCM16)
- **Bytes Per Frame (20ms):** 320 samples * 2 bytes = 640 bytes
- **Throughput:** 32 KB/s

3. VAD & Semantic Chunking Strategy

High latency ki sabse badi wajah bura chunking hota hai. Hum **Hybrid Chunking System** follow karenge:

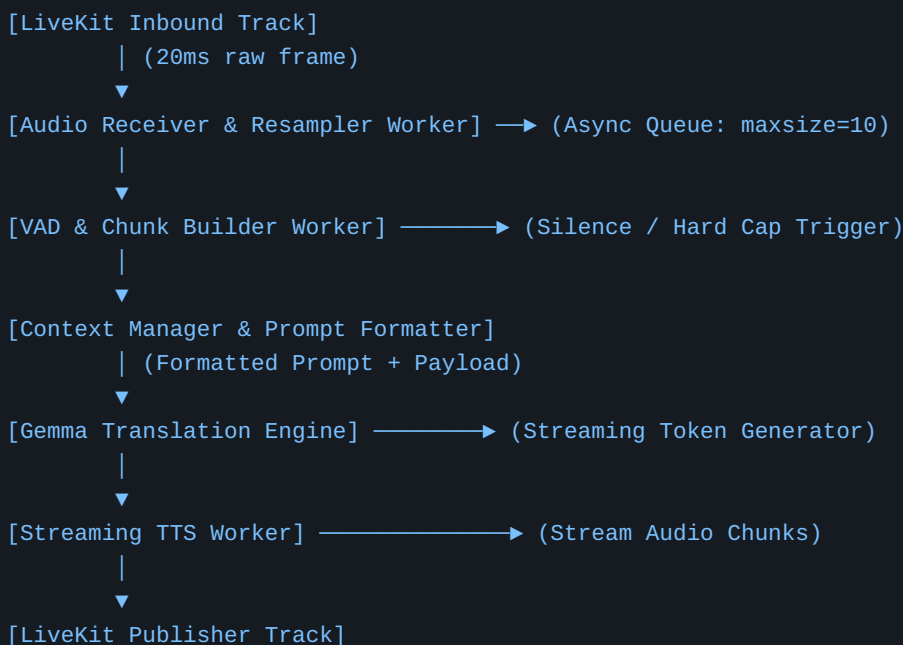
- **Fast VAD Filter:** WebRTC VAD (Aggressiveness Level 2). Real-time me har 20ms frame ko speech/non-speech mark karega.
- **Silence Trigger:** Agar continuous **200ms se 250ms silence** detect hoti hai, to turant Chunk Complete marker trigger hoga.
- **Hard Duration Cap (Max Chunk):** Agar user bina pause liye lagatar bol raha hai, to exact **1.2 seconds** par chunk cutoff hoke Gemma ko bhej diya jayega taaki latency 400ms se upar na jaye.
- **Min Chunk Threshold:** 300ms se chote audio fragments discard honge (noise filtering).

4. Streaming Context Window Management

CRITICAL FOR GEMMA LATENCY: Gemma ka Prefill time linear nahi, quadratic grow karta hai. Single token context delay ko kill kar sakta hai.

- **Context Strategy:** Pure **Time-based Sliding Window**.
- **Max Context Size:** Maximum **3 Seconds OR 40 Tokens** (Last 1-2 turns).
- **Context Purge Policy:** Naya chunk process hote hi sabse purana historical token window se drop hoga. Heavy conversation history ko alag UI state me rakha jayega, model inference prompt me NAHI.

5. Async Multi-Worker Pipeline Architecture



6. Backpressure & Memory Safety Policies

- **Queue Mechanism:** Low-overhead `asyncio.Queue(maxsize=10)` bounded channel.
- **Overflow Policy:** `DROP_OLDEST` (LIFO preference for latency). Agar translation pipeline backup hoti hai, to purana buffered frame drop ho jayega taaki listener ko hamesha current real-time voice mile.

- **Thread Isolation:** VAD/Resampling operations non-blocking C-bindings use karenge. Event loop par CPU spike allow nahi hai.

7. Microsecond Latency Telemetry

Har audio chunk ke saath ek LatencyTelemetry object associate rahega jo ye timestamps log karega:

1. `t0_frame_received` (LiveKit se pehla frame milne ka time)
2. `t1_vad_triggered` (Chunk complete hone ka time)
3. `t2_prompt_ready` (Context payload format hone ka time)
4. `t3_gemma_first_token` (TTFT - Target < 250ms)
5. `t4_tts_first_audio` (TTFA - Target < 350ms)
6. `t5_livekit_published` (End to End - Target < 400ms)